CHAPTER IV

METHODOLOGY

This chapter presents the methodology used in developing the Smart Water Monitoring Consumption System. Following the sequence of the supplied Chapter 4 reference, it discusses requirements and use cases, system design and user interfaces, the entity relationship diagram, the data dictionary, development and testing, the implementation plan, encountered challenges, and implementation results. All technical statements are based on the implemented Laravel application, ESP32 firmware, database migrations, routes, services, views, and automated tests of the Smart Water project.

4.1 Requirements Analysis and Requirements Documentation

Requirements analysis identified three principal actors: the municipal administrator, the household account holder, and the commissioned smart meter. The administrator manages household records, municipal meters, water-rate configuration, billing statements, leak incidents, notifications, analytics, reports, and audit information. The household user reviews live and historical consumption, estimated billing, leak status, notifications, and account settings. The smart meter measures flow-sensor pulses and submits authenticated interval readings to the server.

4.1.1 Functional Requirements

ID	Requirement	Verified Project Behavior
FR-01	User and access management	Administrator creates and manages household accounts; public registration is restricted; account status governs access.
FR-02	Municipal device lifecycle	Register, assign, activate, rotate credentials, reassign, suspend, and decommission meters.

ID	Requirement	Verified Project Behavior
FR-03	Telemetry ingestion	Accept authenticated interval readings containing a stable reading identifier, flow rate, liters used, meter state, and device diagnostics.
FR-04	Consumption monitoring	Display live readings, daily/weekly/monthly history, summaries, comparisons, and availability states.
FR-05	Estimated billing	Apply the latest applicable water rate, fixed charge, subsidy, and billable consumption to estimates and monthly statements.
FR-06	Leak and threshold detection	Evaluate server-configured thresholds, notify affected users, and support an administrator resolution workflow.
FR-07	Notifications	Create, read, archive, retry, and remove household or administrative notices according to role and ownership.
FR-08	Reports and exports	Produce consumption, billing, leak, notification, and operational reports with filtering and export support.
FR-09	Audit and analytics	Record administrative/security activity and present household- or administrator-scoped analytics.
FR-10	Account maintenance	Allow authenticated users to update permitted profile fields and change passwords.

4.1.2 Non-Functional Requirements

Quality Attribute	System Response
Security	Laravel Sanctum tokens, role middleware, scoped device credentials, hashed one-time activation codes, token revocation, and audit redaction.
Reliability	Stable reading identifiers prevent double counting; unsent meter intervals are retained in ESP32 Preferences until acknowledged.
Performance	The meter uploads at three-second intervals and the web interface is designed to refresh live information at the same cadence.

Quality Attribute	System Response
Data integrity	Foreign keys, validation, unique identifiers, server-owned household/device associations, precision-preserving numeric fields, and historical charge snapshots.
Usability	Responsive administrator and household workspaces, accessible labels, keyboard-oriented controls, role identification, and light/dark theme support.
Maintainability	Controllers are grouped by feature; models represent domain entities; services isolate billing, analytics, consumption, and audit logic.
Portability	The web application supports SQLite or MySQL, while the field device targets ESP32-S3/M5Stack CoreS3 hardware.

4.1.3 Use-Case Description

Use Case	Actor	Precondition	Main Flow	Result
Manage household accounts	Administrator	Authenticated administrator	Create, search, filter, edit, suspend, deactivate, or archive a household account.	A role-scoped household record with a standardized account number.
Commission a smart meter	Administrator + Device	Registered and eligible household	Register the asset, issue a one-time code, activate from the ESP32, and bind the hardware identity.	Active scoped device credential and assigned meter.
Upload a reading	Smart meter	Active assigned device and valid token	Measure pulses, calculate interval values, submit JSON, validate identity, and save once.	Owned water reading and updated server total.
Review consumption	Household	Authenticated active account	Open live monitoring or history; select a period; retrieve scoped readings and summaries.	Current flow, usage totals, trends, and availability state.
Generate billing statement	Administrator	Valid rate and billable readings	Select the billing month, aggregate consumption, apply charge rules, and save a snapshot.	Finalized statement with status and balances.

Use Case	Actor	Precondition	Main Flow	Result
Resolve a leak alert	Administrator	Leak incident exists	Review household and reading evidence, notify, classify, resolve, mark false alarm, or reopen.	Updated incident state and traceable response.

4.1.4 Use-Case Diagram

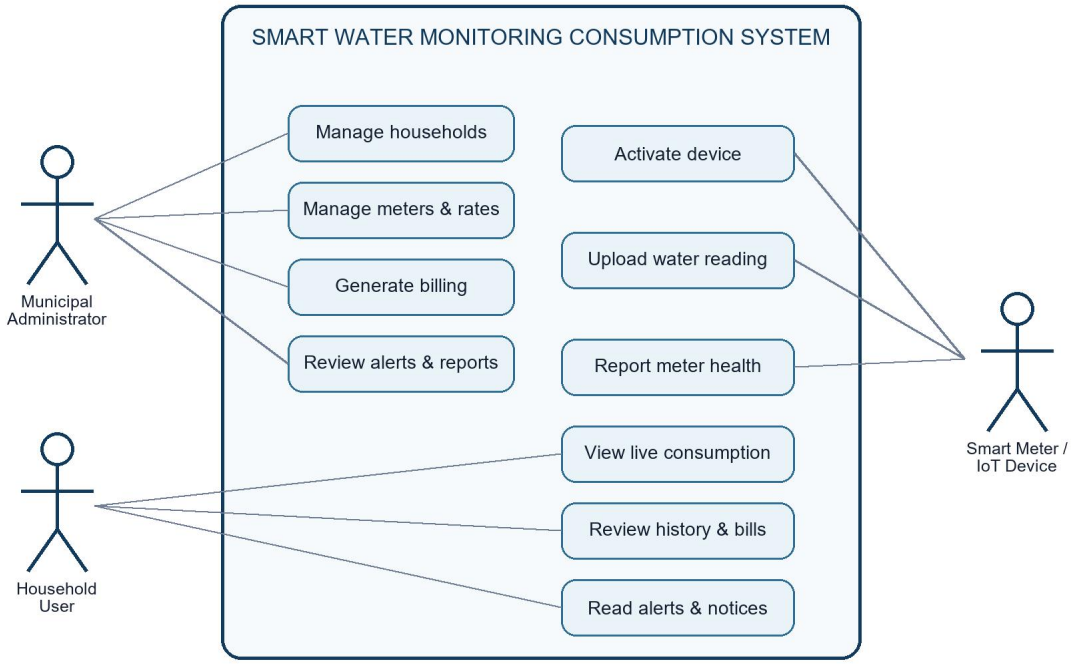


Figure 4.1. Use-case overview of the Smart Water Monitoring Consumption System.

4.2 System Design, Software, and Processes

The system uses an Internet of Things client-server design. The physical meter performs measurement and short-term persistence, while the Laravel server remains authoritative for identity, ownership, cumulative usage, billing rules, threshold evaluation, and record storage. The browser interfaces consume protected JSON endpoints and present role-specific workspaces.

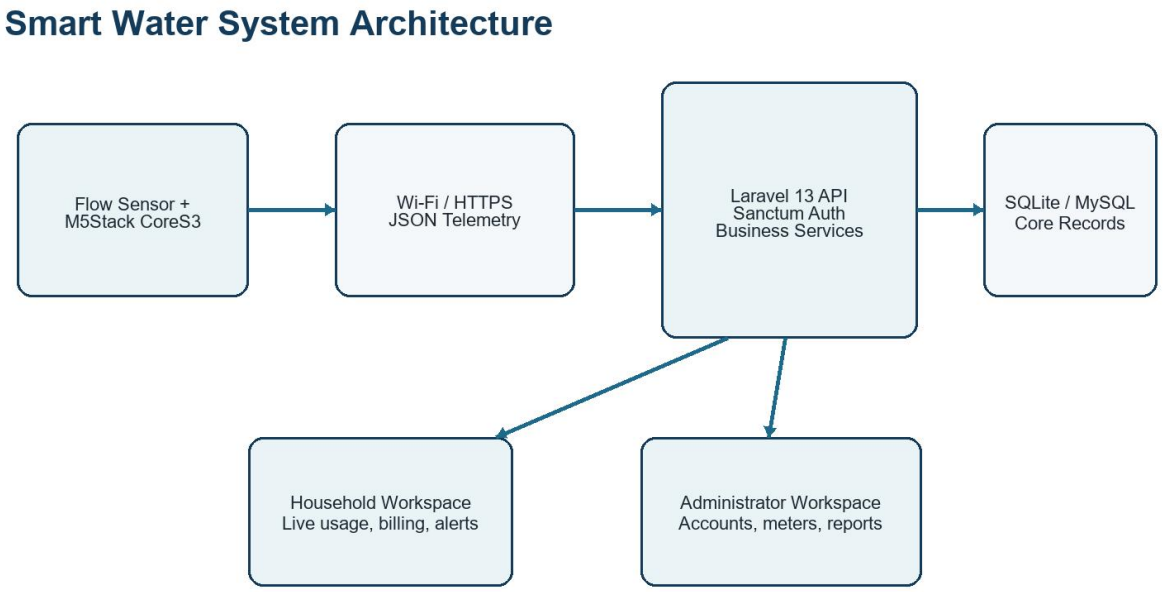


Figure 4.2. Smart Water Monitoring Consumption System architecture.

4.2.1 Telemetry and Billing Process

Commissioning begins in the administrator workspace. The device exchanges its UUID, expiring activation code, hardware ID, and firmware version for a scoped credential. Each reading includes a stable reading_id derived from the registered device identity and local sequence. If transmission fails, the pending interval remains stored in the device. When the API acknowledges the reading, the local sequence advances and the interval accumulator is cleared. The server validates ownership and computes authoritative totals before exposing information to users.

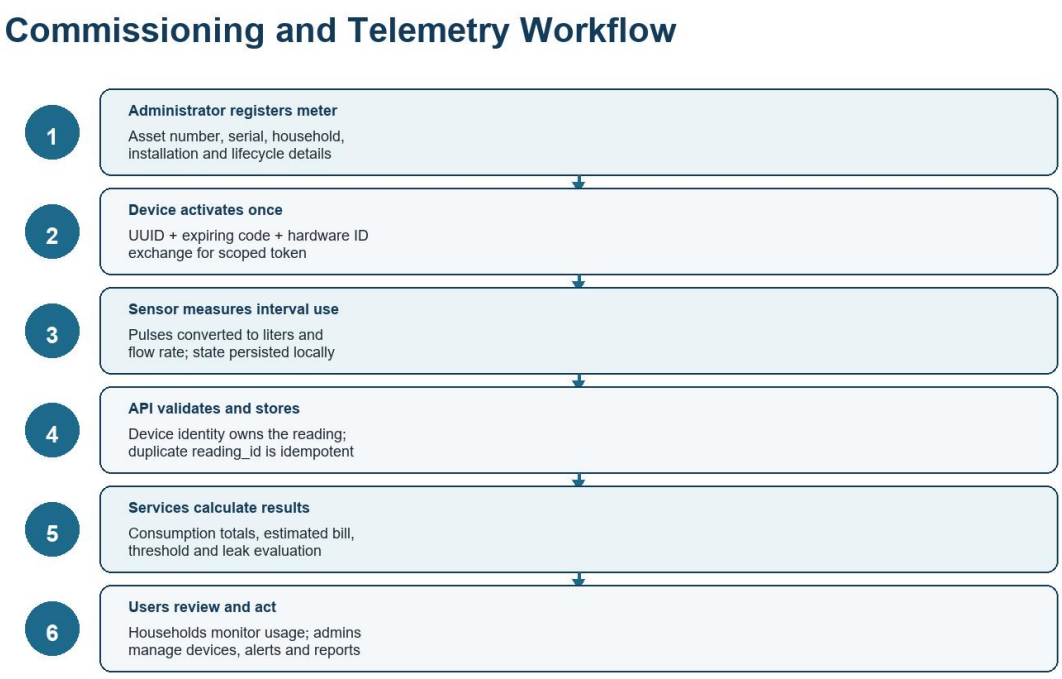


Figure 4.3. Device commissioning, telemetry, calculation, and user-response workflow.

4.2.2 Technology Used

Layer	Technology	Purpose
Backend	PHP 8.3+; Laravel Framework 13.8; Laravel Sanctum 4.0	Routing, validation, authentication, API controllers, services, reports, and persistence.
Frontend	Blade, JavaScript, Tailwind CSS 4, Vite 8	Responsive household and administrator workspaces, charts, filters, forms, and live refresh.
Database	SQLite or MySQL	Users, meters, readings, rates, billing statements, notifications, leak alerts, audit logs, sessions, and tokens.
Embedded device	ESP32-S3 / M5Stack CoreS3; M5Unified; Wi-Fi; HTTPClient; Preferences	Pulse measurement, device display, activation, reliable telemetry, and local pending-state storage.
Reporting	Dompdf 3.1.2 and CSV export endpoints	Printable operational reports and downloadable tabular records.
Testing	PHPUnit 12.5.12 through Laravel test runner	Feature verification for access control, telemetry, billing, alerts, analytics, devices, and interfaces.

4.2.3 Output and User Interface Design

The output and user interface design separates household monitoring from municipal administration. Household screens prioritize current flow, consumption history, billing estimates, leak status, notifications, and account settings. Administrator screens prioritize customers, devices, rates, statements, consumption, leak incidents, reports, notifications, analytics, and audit information. The responsive Blade layouts maintain consistent navigation while preserving role-based access boundaries.

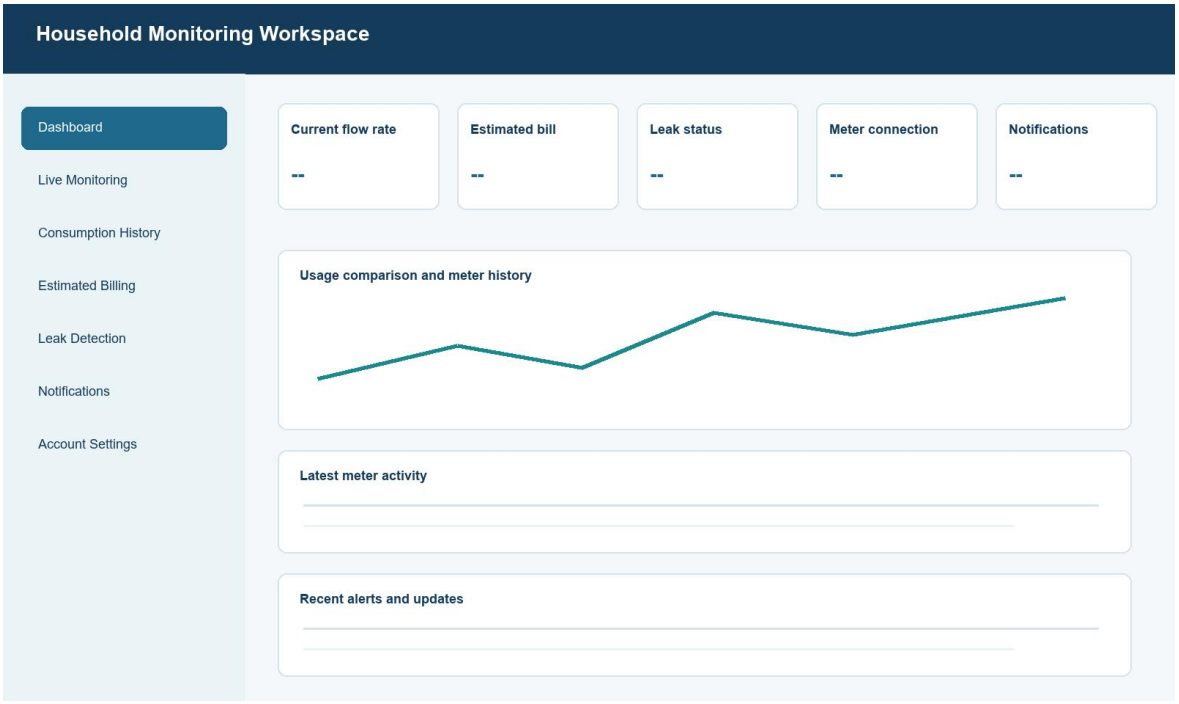


Figure 4.4. Household workspace interface map.

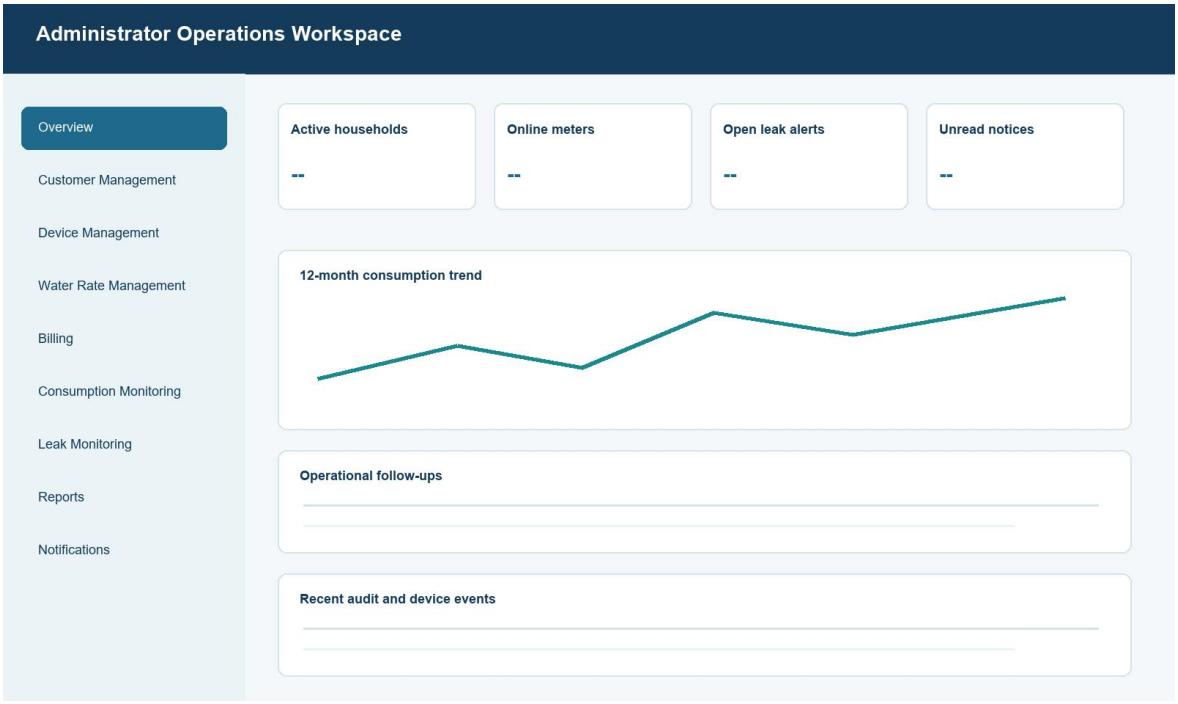


Figure 4.5. Administrator workspace interface map.

User Interface	Primary Outputs	Design Purpose
Household dashboard	Current flow, usage, estimated bill, meter status, leak state, and recent notices	Present essential account information at a glance.
Live monitoring and history	Recent readings, charts, period totals, summaries, and availability states	Support near-real-time review and time-based consumption analysis.
Billing and alerts	Rate breakdown, statements, leak status, thresholds, and notifications	Make charges and exceptional conditions understandable and actionable.
Administrator workspace	Household, meter, rate, billing, consumption, alert, report, and audit records	Centralize municipal operations while maintaining traceability.

4.3 Entity Relationship Diagram

The principal database relationships connect household users to their assigned devices, readings, statements, notifications, and leak incidents. A device may be temporarily unassigned during municipal inventory or maintenance, but an uploaded reading is tied to both the authenticated device and its current household. Billing configuration can be municipal or user-specific, and finalized statements preserve the charge values applied at generation time.

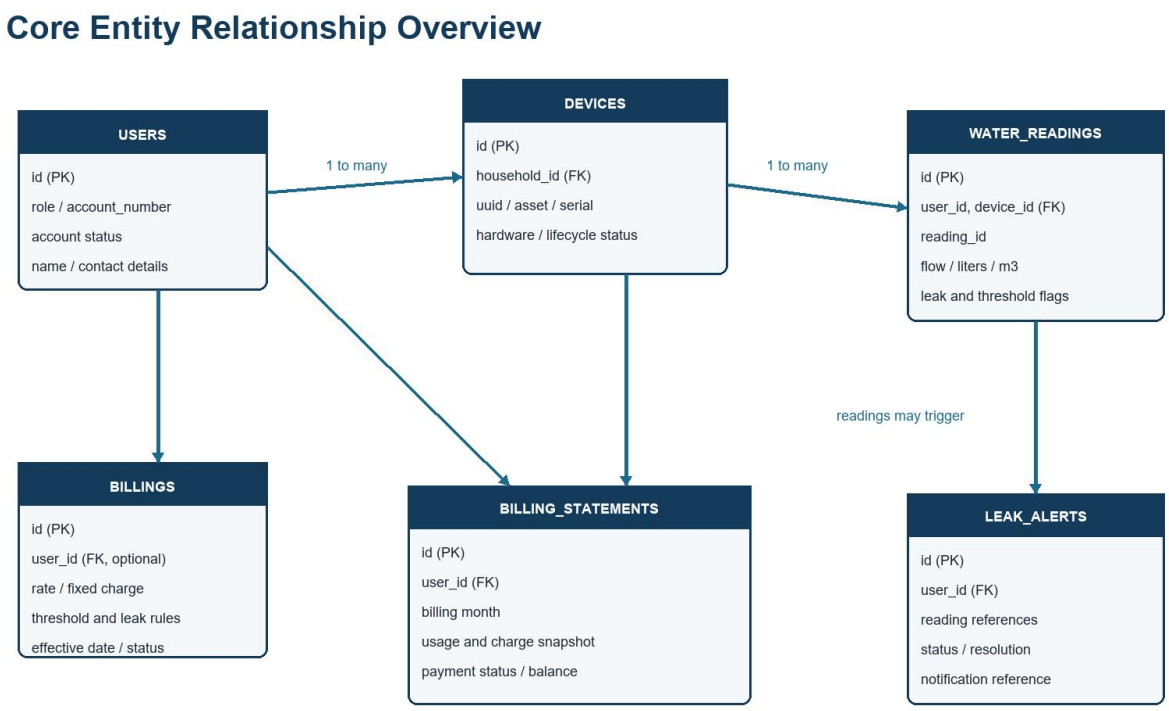


Figure 4.6. Core entity relationship overview (operational fields summarized).

4.4 Data Dictionary

The following data dictionary summarizes the operational tables most directly involved in monitoring, billing, alerting, and administration. Framework-support tables for cache, jobs, sessions, password resets, and personal access tokens remain part of the implementation but are omitted from the main discussion for clarity.

Table	Important Fields	Purpose
users	id; name; email; role; account_number; address; phone; is_active; account_status; created_by_admin; timestamps	Household and administrator identities, contact details, and access state.
devices	id; household_id; device_uuid; asset_number; manufacturer_serial; hardware_id; firmware_version; installation_date; status; last_seen_at; activation metadata	Municipal meter registry, assignment, identity binding, health, and lifecycle.
water_readings	id; user_id; device_id; reading_id; flow_rate; liters_used; cubic_meter; estimated_bill; rate; threshold and leak flags; connectivity fields; recorded_at	Time-stamped, device-owned interval telemetry and derived monitoring state.

Table	Important Fields	Purpose
billings	id; user_id; water_rate; fixed_monthly_charge; subsidy_deduction; currency; effective_date; threshold_liters; leak rules; is_active	Effective-dated municipal or household billing and monitoring configuration.
billing_statements	id; user_id; billing_month; opening_balance; total_liters; total_cubic_meter; water_rate; fixed charge; subsidy; current_charge; amount_paid; remaining_balance; status	Historical monthly statement and payment-state snapshot.
notifications	id; user_id; created_by; type; title; message; read state; batch; delivery status/error; sent/scheduled/archived timestamps	Household notices, administrator announcements, delivery lifecycle, and read status.
leak_alerts	id; user_id; first/latest reading IDs; notification_id; status; detected/resolved timestamps; resolution notes; last notified date	Operational incident record supporting notification, classification, resolution, false-alarm handling, and reopening.
audit_logs	id; user_id; action; entity type/id; metadata; IP address; user agent; request ID; created_at	Trace of administrative and security-sensitive activity with contextual metadata.

4.5 Development and Testing

Development is organized around feature-specific API controllers and domain services. WaterConsumptionService and HouseholdConsumptionService handle aggregation and availability semantics; BillingStatementService preserves monthly charge logic; AnalyticsService produces range-based summaries; and the embedded firmware handles acquisition, display, activation, and delivery reliability. Verification combines automated feature tests with a production frontend build.

Verification	Coverage / Output	Status	Duration	Observation
Laravel automated test suite	138 tests; 885 assertions	137 passed; 1 failed	18.017 seconds	The failing case expected one record but received two in an administrator consumption-history response.
Frontend production build	Vite 8.0.16; 4 modules transformed	Passed	6.75 seconds	CSS, boot JavaScript, main JavaScript, and manifest were produced successfully.

Current verification status: The project is substantially covered by automated tests, but the suite is not fully green as of 24 August 2026. One consumption-monitoring test must be investigated before claiming complete regression-test acceptance.

4.5.1 Major Test Areas

- Authentication and account restrictions, including blocked, suspended, archived, deleted, and unauthorized users.
- Municipal device registration, activation, hardware binding, credential rotation, reassignment, health, and decommissioning.
- Telemetry validation, device ownership, idempotent reading identifiers, server-authoritative totals, and precision.
- Household live readings, time-zone-aware periods, cumulative-reset handling, missing-data semantics, and billing estimates.
- Water-rate configuration, fixed monthly charges, statement generation, status changes, and historical billing fields.
- Leak alert creation, notification, resolution, false-alarm handling, reopening, and role protection.
- Administrator reporting, analytics, customer management, exports, notification lifecycle, and interface accessibility contracts.

4.6 Implementation Plan

Phase	Owner	Activity	Expected Output
1. Environment preparation	Municipal IT / Developers	Configure PHP, database, application key, mail/queue behavior, timezone, LAN URL, and production assets.	Environment checklist and successful health endpoint.
2. Database deployment	Developers	Back up the target database, run migrations and controlled seeders, and verify indexes and foreign keys.	Current schema and verified baseline records.
3. Administrator setup	System administrator	Create authorized staff accounts, set water rates and alert rules, and verify audit logging.	Operational administrative workspace.

Phase	Owner	Activity	Expected Output
4. Household onboarding	Authorized staff	Create household records with required address, contact details, role, status, and account number.	Active household accounts ready for assignment.
5. Meter commissioning	Technician + Administrator	Register each meter, record asset details, assign household, activate once, and verify last-seen health.	Bound device and successful telemetry.
6. Pilot verification	Project team	Compare measured volume with a calibrated reference, verify network recovery, review alerts, and reconcile billing estimates.	Signed pilot acceptance record.
7. Production rollout	Municipality	Deploy in phases, train staff and households, monitor audit/leak/device dashboards, and maintain backups.	Controlled operational adoption.

4.6.1 Implementation Issues and Challenges

Issue	Effect	Mitigation
Sensor calibration	Pulse-to-liter conversion varies by sensor, pipe conditions, pressure, and installation.	Perform volumetric calibration per installed meter and retain calibration records.
Network interruption	LAN or Wi-Fi loss can delay near-real-time visibility.	Use the firmware pending-reading state, stable IDs, health windows, and field connectivity checks.
Duplicate or noisy pulses	Electrical noise may inflate flow readings.	Use minimum pulse intervals, abnormal-burst rejection, and site validation.
Credential handling	Activation codes or tokens may be exposed during commissioning.	Issue short-lived single-use codes, remove codes after activation, rotate credentials, and restrict administrator access.
Billing configuration	Incorrect effective dates or rate settings can affect estimates and statements.	Require authorized review, retain charge snapshots, and reconcile pilot statements before rollout.
Test-suite regression	One current consumption-history test has a record-count mismatch.	Isolate the query/date fixture, correct the root cause, and rerun the complete suite before acceptance.

4.7 System Implementation and Results

The implemented application provides a complete path from a registered physical meter to role-scoped monitoring and administrative action. The meter reports measured intervals; the API validates device identity; the database preserves raw and derived values; services calculate consumption and estimated billing; and the user interfaces expose monitoring, leak, notification, billing, analytics, and reporting functions. The successful production asset build and 137 passing automated tests demonstrate broad functional coverage, while the remaining failing test identifies a specific acceptance item rather than an undocumented weakness.

Interpretation: The implementation is suitable for continued pilot testing and controlled demonstration. Production acceptance should follow calibration, backup, security, network-recovery, and full regression-test sign-off.